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Brains of Vespertilionids

I. Subfamily characteristics

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Abstract

Macromorphology and encephalization (EI) of brains were compared in 58 Vespertilionid species, brain composition in 36 species: 46 or 27 species of Vespertilioninae, 8 or 5 species of Miniopterinae, 2 species of Kerivoulinae, and 2 species of Nyctophilinae. Subfamily differences were found in the extent of the cover of the mesencephalon. It is nearly fully covered in *Kerivoula papillosa* (Kerivoulinae), at least half covered (by the cerebellum) in Miniopterinae, and free (completely or nearly so) in Nyctophilinae and Vespertilioninae. In relative brain size, the Kerivoulinae are highest (average EI = 130), followed by the Miniopterinae (111), Nyctophilinae (102) and Vespertilioninae (95). The higher encephalization of Kerivoulinae and Miniopterinae is accompanied by a marked increase of relative size in cerebellum and striatum, and in Kerivoulinae, in hippocampus and neo-cortex as well.

Key words: Brain – Macromorphology – Encephalization – Morphometry – Brain composition – Vespertilionid Bats

Introduction

A reliable instrument for assessing relative brain size in Chiroptera is the Vespertilionid regression line after exclusion of the two *Tylonycteris* species (STEPHAN et al. 1981 b). The Vespertilionids were chosen because they: 1. have, on the average, the lowest relative brain size, i. e., are the least encephalized of all Chiroptera families, 2. are represented in our material by a large number of species, 3. have a wide range of body weights, which stabilizes the slope of the regression line, and 4. have a slope similar to that of the Pteropodids, which have the largest range of body weights and a large number of species.

Tylonycteris spp. were excluded since their encephalization indices were distinctly below those of the other Vespertilionids. Their inclusion would have enlarged the gap between the poorly encephalized Vespertilionids and other bat families. A still lower encephalization ($EI = 90$) than in the average Vespertilionid ($EI = 100$) was found by STEPHAN et al. (1981b) for *Natalus tumidirostris* (Natalidae) based on values taken from the data of PIRLOT and POTTIER (1977). Since publication of their report, we collected 10 individuals in Trinidad and found a standard BoW of 6.9 g (instead of their 3.7 g) and a BrW of 245 mg (instead of 100 mg). The resulting EI is 144, which is even higher than the maximum found in Vespertilionids.

The importance of the Vespertilionids as the reference group for brain comparisons in Chiroptera makes it desirable to know more about their brains and the variability within this family. This paper (Part I) focusses on brain characteristics (similarities and variability) of the various Vespertilionid subfamilies. In other papers, Vespertilioninae (Part II) and comparative architectonics of the hippocampus in Vespertilionids (Part III) are examined in more detail.

Material and methods

Investigations on the relative size of the *total brains* were based on 336 specimens from 58 species of Vespertilionids: 323 specimens are from our own collection, and the BoWs and BrWs of 13 individuals were taken from the literature; their origin is given in STEPHAN et al. (1981b). The Vespertilioninae are represented by 259 specimens from 46 species, the Miniopterinae by 67 specimens from 8 species or subspecies, the Kerivoulinae by 2 specimens from 2 species and the Nyctophilinae by 8 specimens from 2 species. We have no material from Murinae and Tomopeatinae. The encephalization of a given species is determined from standardized brain and body weights (BrW and BoW), which take into account 1. possibly different numbers of males and females (an inconvenience when sex differences exist), 2. net BoWs of gravid females (obtained by subtraction of uterus weights), and 3. young animals (exclusion when clearly lower than the data on adults). The standards are given in Table 1; the method for determining the standards is shown for *Tylonycteris* in Part II of this series of reports. The standards of the BrWs are plotted against the standards of the BoWs in a double logarithmic scale (Fig. 4). The least-square regression line for the 56 non-*Tylonycteris* Vespertilionid species was determined and used as reference line. Any value on this line was considered to represent the average Vespertilionid. The vertical distances from the reference line express the degree of deviation of the BrW of a given species from those of the isoponderous typical Vespertilionid. The reference line itself has, of course, a distance of 0, the antilog of which is 1 (or 100%). The deviations from the Vespertilionid regression line are expressed by encephalization indices (EIs) and given in Table 1 in percentage form. The thinner parallel lines drawn in Fig. 4 in distances of 0.301 (i.e., the logarithm of 2) indicate when a doubling ($2\times$ or 200%) or a halving ($\frac{1}{2}\times$ or 50%) of the baseline encephalization value is reached.

The comparisons of *brain shape* are based on the 323 Vespertilionid brains in our collection.

The comparisons of *brain composition* are based on 79 serially sectioned brains from 36 species. The Vespertilioninae are represented in this material by 59 brains of 27 species, the Miniopterinae by 13 brains of 5 species or subspecies, the Nyctophilinae by 4 brains of 2 species, and the Kerivoulinae by 3 brains of 2 species. Serial sections of one of the *Kerivoula papillosa* brains were kindly made available by P. PIRLOT (Montréal). The methods for volume determination are described in detail in STEPHAN et al. (1981a). All volumes were corrected to fresh brain values. For the comparisons, size indices were determined by the same method as described for the total brain. Instead of the BrWs, the average volumes of the measured structures (listed in Table 2) were plotted against BoWs. For each structure, the regression line of the non-*Tylonycteris* Vespertilionid species was determined and used as the reference baseline for calculating the size indices (SI). This is shown for the striatum in Fig. 5.

The percentage values of the brain components relative to the total net brain volume (NBV) or total telencephalon (TEL) can easily be determined from the volume data of Table 2 and are therefore not given.

In Part II (Figs. 4 and 5), some histological sections are shown, and the borders of brain components are marked.

List of abbreviations

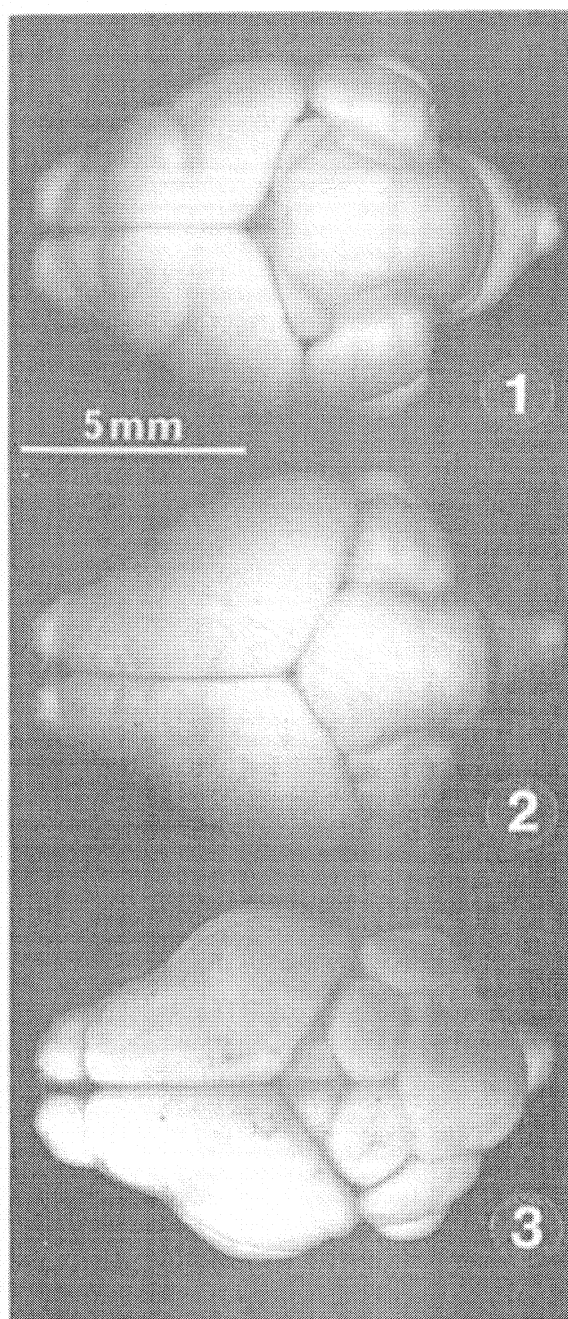
BOL	bulbus olfactorius
BoW	body weight
BrW	brain weight
CER	cerebellum
CV	coefficient of variation ¹
DIE	diencephalon
EI	encephalization index
HIP	hippocampus
MES	mesencephalon
NBV	net brain volume
n	number of specimens
N	number of species
NEO	neocortex
OBL	medulla oblongata
PAL	paleocortex + amygdala
STR	striatum
SCH	schizocortex
SEP	septum
SI	size index
TEL	telencephalon

Results

Macromorphology

The brains of *Miniopterinae* differ clearly from those of other subfamilies. Their cerebral hemispheres have two distinct sulci (Fig. 1), which in all other Vespertilionid subfamilies are at the most merely outlined or foreshadowed (Figs. 2, 3). The larger sulcus is in the position of the sulcus lateralis cerebri (Sylvii) of other mammals, separating the frontal 'lobe' from the more caudal parts of the hemisphere, and the smaller sulcus runs transversely slightly behind the frontal pole (sulcus rostralis). Similar sulci were described by STEPHAN (1977) for the *Mormoopidae* (= *Chilonycteriinae* in that paper).

Another brain characteristic, which may be markedly different in various subfamilies, is the extent to which the mesencephalic tectum is covered (see also SCHNEIDER 1957; STARCK 1963), mainly by the cerebellar vermis. Based on the extent of cover of the median longitudinal fissure of the tectum, four phases can readily be differentiated: 1. the fissure is visible over its whole or most of its extent, both between caudal and rostral colliculi; 2. the cerebellar vermis covers the longitudinal fissure of the caudal colliculi, whereas the rostral colliculi and their longitudinal



Figs. 1-3. Dorsal view of brains of *Miniopterus schreibersi* (Fig. 1), *Kerivoula papillosa* (Fig. 2), and *Nyctophilus timoriensis* (Fig. 3). (Linear enlargement 5 ×)

¹ Standard error of the mean in percentage of the mean.

fissure remain visible; 3. medio-caudal parts of the rostral colliculi are also covered; and 4. the whole tectum is covered. The Miniopterinae, and only these, belong to phase 3 (Fig. 1). Phase 4 is occupied by *Kerivoula papillosa* only (Fig. 2), phase 2 by *K. phalaena*, *Chalinolobus* spp. and *Glauconycteris* spp., and phase 1 by most of the Vespertilioninae and the Nyctophilinae (Fig. 3). *Nyctalus*, *Lasius* and several species of *Myotis* and

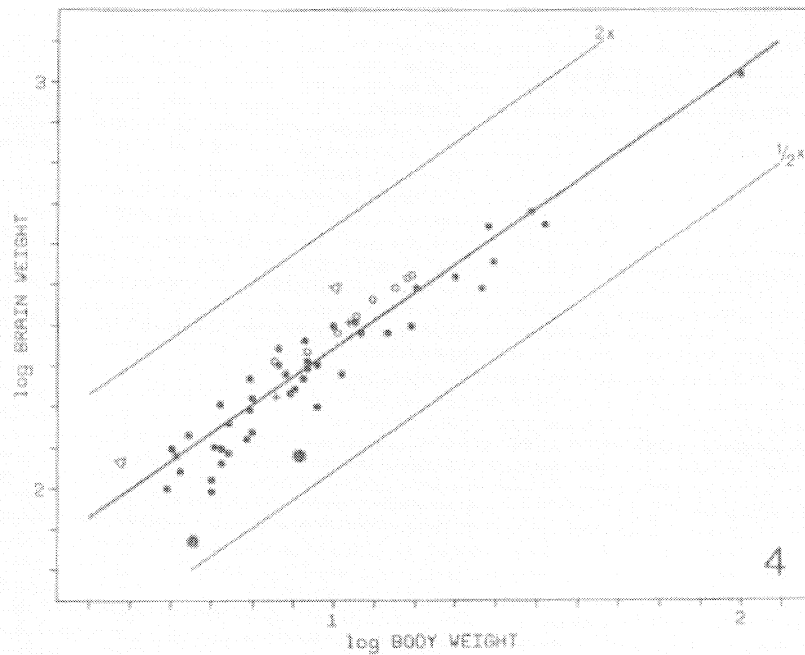


Fig. 4. Brain weight (BrW in milligrams) plotted against body weight (BoW in grams) in a double logarithmic scale. The reference baseline (thick) is the least-square regression line of the 56 non-*Tylonycteris* Vespertilionid species from which BrWs and BoWs were available. The equation is: $\log \text{BrW} = 1.656 + 0.686 \cdot \log \text{BoW}$. The thinner parallel lines are at distances of 0.301, i. e., in multiples and fractions of 2; they correspond to the encephalization indices 2 (or 200%) and $\frac{1}{2}$ (or 50%).
 • non-*Tylonycteris* Vespertilioninae; • *Tylonycteris* spp.; o Miniopterinae; ∇ Kerivoulinae; + Nyctophilinae

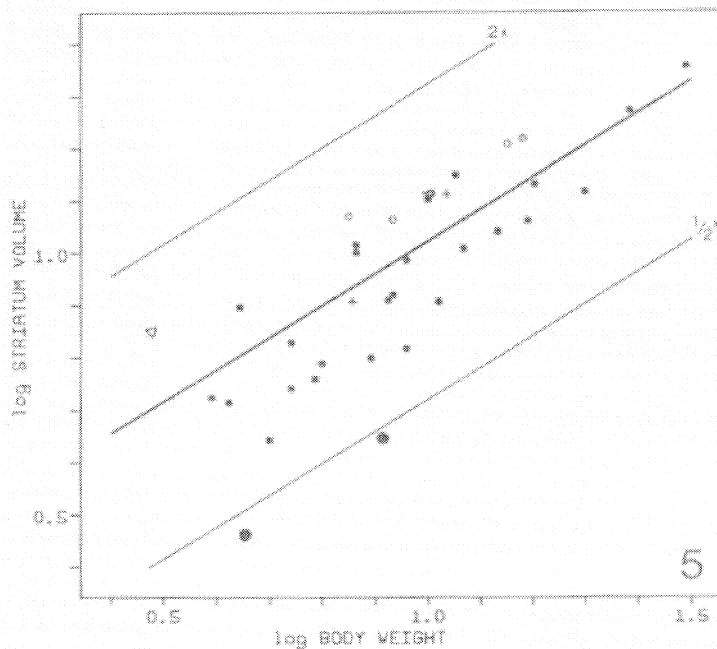


Fig. 5. Striatum volume (in mm^3) plotted against BoW in a double logarithmic scale. The reference baseline (thick) is the least-square regression line of the 34 non-*Tylonycteris* Vespertilionid species for which we have serially sectioned brains. The equation is: $\log \text{striatum volume} = 0.431 + 0.608 \cdot \log \text{BoW}$. For further explanations and abbreviations, see Fig. 4.

Pipistrellus are intermediate between phases 1 and 2. Thus, the tectum is largely uncovered in the Vespertilioninae and Nyctophilinae, partly covered in the Miniopterinae, and nearly completely covered in *Kerivoula papillosa* (but distinctly less so in *K. phalaena*).

Differences in the general brain shape are not related to specific subfamilies. The greatest variation in all the Vespertilionid brains is found in the Vespertilioninae, between the extremely flat brains of *Tylonycteris* (Figs. 1 and 4 in Part II) and the high and very short brains of *Glauconycteris*, *Chalinolobus*, and *Lasiurus* (Fig. 3 in Part II). The general Vespertilionid type is represented by the majority of Vespertilioninae and by Miniopterinae, Kerivoulinae, and *Nyctophilus timoriensis*. A slight flattening of the brain was found in *Glischropus* and *Nyctophilus geoffroyi*, but to a lesser extent than in *Tylonycteris*.

Encephalization

In a former paper (STEPHAN et al. 1981b), a slope of 0.684 was found for the regression line of the Vespertilionid brains after exclusion of the two *Tylonycteris* species (N = 43). Since that study, the material was enlarged by 13 species, and the slope of the new regression line (without *Tylonycteris* spp., N = 56) is now 0.686 (thick line in Fig. 4).

It should be noted that the average encephalization of the 56 non-*Tylonycteris* Vespertilionids is by definition = 1 or 100 (%).

Table 1

Standard values of body weights (BoW in grams), brain weights (BrW in milligrams) and encephalization indices (EIs, $\times 100$) of 58 Vespertilionid species

Code	Subfamilies and species	n	BoW	BrW	EI
Vespertilioninae					
2377	<i>Myotis adversus</i>	5	11.3	255	107
2381	<i>albescent</i>	3	4.0	125	107
2399	<i>bechsteini</i>	2	10.0	250	114
2403	<i>bocagei</i>	6	7.3	200	113
2419	<i>dasyneura</i>	1	16.0	310	102
2421	<i>daubentonii</i>	1	8.5	230	117
2439	<i>fortidens</i>	9	4.4	135	108
2473	<i>lucifugus</i>	12	8.0	175	93
2487	<i>muricola</i>	10	5.1	126	91
2489	<i>myotis</i>	7	31.0	480	101
2491	<i>mystacinus</i>	3	5.25	160	113
2495	<i>nattereri</i>	1	7.3	220	124
2497	<i>nigricans</i>	12	4.4	135	108
2498	<i>nigricans</i> ssp.	7	6.2	155	98
2591	<i>Pipistrellus crassulus</i>	4	6.1	132	84
2601	<i>imbricatus</i>	15	6.3	137	86
2631	<i>nannus</i>	17	3.9	100	87
2652	<i>spec?</i>	2	4.1	120	101
2655	<i>subflavus</i>	2	5.3	125	98
2659	<i>tenuis</i>	12	5.3	115	81
2660	<i>sp?</i>	1	6.3	166	104
2671	<i>Nyctalus noctula</i>	1	25.0	360	87
2672	<i>stenopterus</i>	5	15.5	250	84
2675	<i>Glischropus tylops</i>	3	5.0	98	72
2681	<i>Eptesicus brasiliensis</i>	13	9.1	200	97
2695	<i>flavescens</i>	2	10.5	190	84
2699	<i>fuscus</i>	2	13.6	240	88
2700	<i>gaumeri</i>	4	7.6	190	104
2712	<i>melanopterus</i>	5	8.6	196	99
2723	<i>pumilus</i>	4	5.5	122	84
2731	<i>serotinus</i>	1	23.4	310	79
2747	<i>Vespertilio murinus</i>	1	33.3	445	89
2773	<i>Tylonycteris pachypus</i>	15	4.5	74	59
2775	<i>robustula</i>	19	8.2	120	62
2796	<i>Glauconycteris humeralis</i>	3	5.5	144	99
2799	<i>poensis</i>	1	8.6	205	103
2807	<i>Chalinolobus gouldi</i>	3	11.7	240	98
2809	<i>morio</i>	5	8.4	185	95
2831	<i>Nycticeius balstoni</i>	1	9.1	158	77
2843	<i>Rhogeessa parvula</i>	7	4.2	110	91
2845	<i>tumida</i>	3	5.0	105	77
2857	<i>Scotophilus dinganii</i>	13	24.3	440	109
2861	<i>kuhli</i>	13	20.0	330	93
2865	<i>nigrita</i>	1	101	1046	97
2877	<i>Lasiurus borealis</i>	1	7.8	170	92
2889	<i>Barbastella barbastellus</i>	1	6.2	185	117
Miniopterinae					
2907	<i>Miniopterus australis</i>	6	7.1	205	118
2911	<i>inflatus</i>	8	15.2	328	112
2913	<i>medius</i>	10	8.6	216	109
2918	<i>schreibersi</i> ssp.	10	11.4	263	109
2919	<i>a.schreibersi</i>	6	12.5	290	113
2920	<i>a.arenarius</i>	12	10.2	240	108
2921	<i>a.blepotis</i>	7	14.2	310	111
2922	<i>tristis</i>	8	15.6	333	112
Kerivoulinae					
2967	<i>Kerivoula papillosa</i>	1	10.1	310	140
2971	<i>phalaena</i>	1	3.0	116	121
Nyctophilinae					
2999	<i>Nyctophilus geoffroyi</i>	5	7.2	167	95
3005	<i>timoriensis</i>	3	10.9	254	109

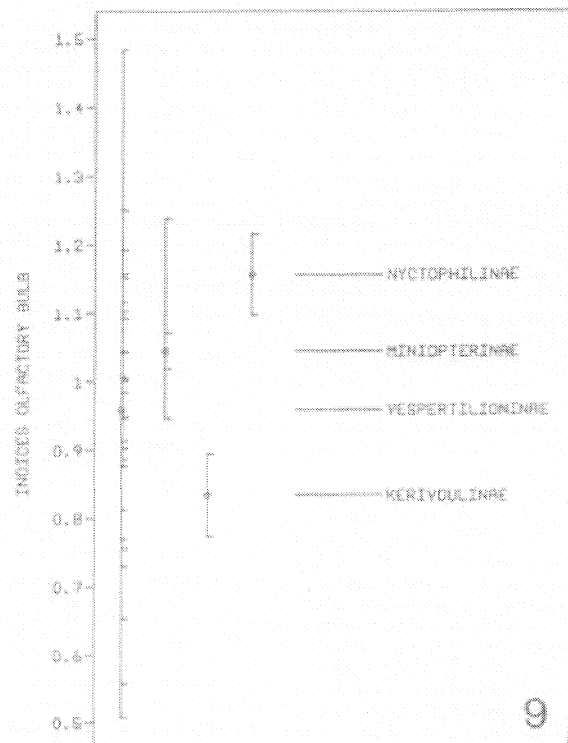
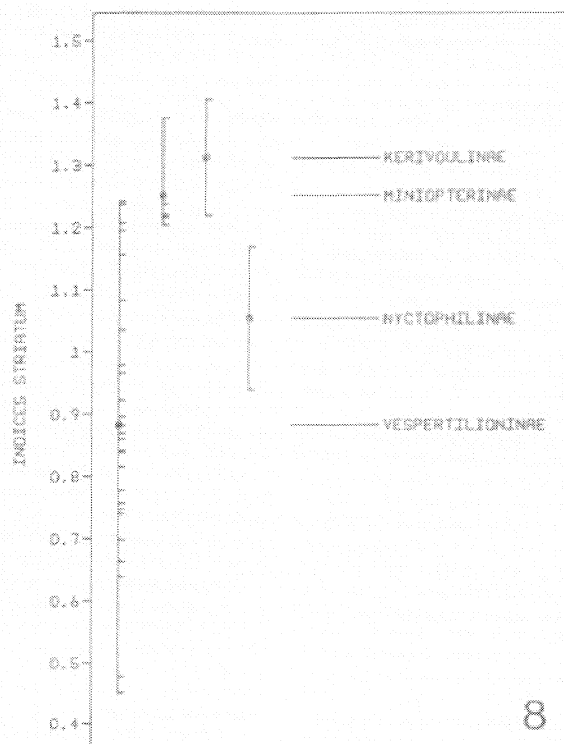
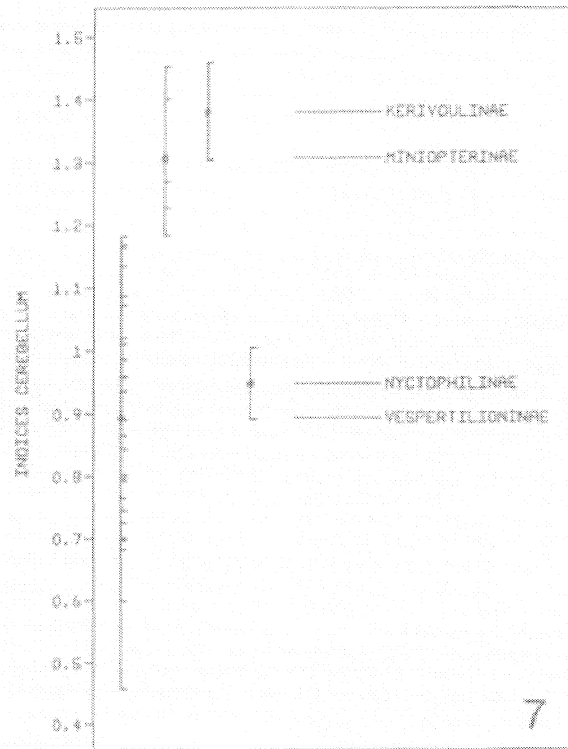
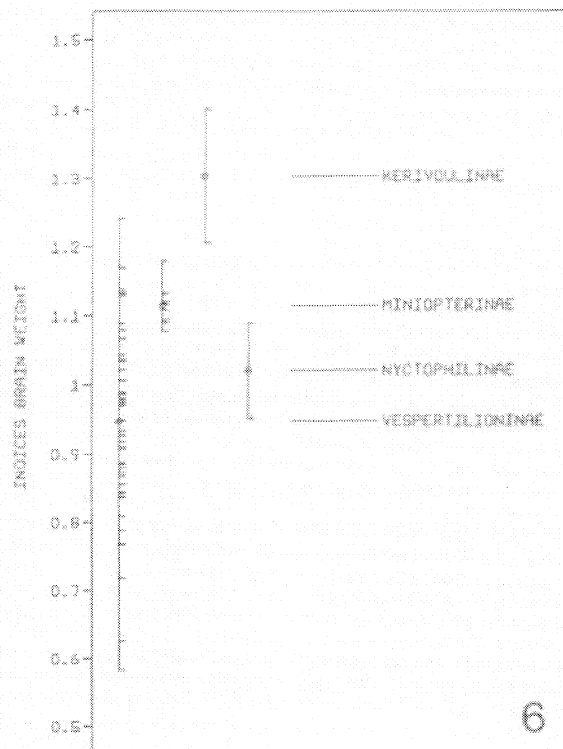


Fig. 6. Encephalization indices of the four Vespertilionid subfamilies available. The EIs are numerical values of the distances from the reference line in Fig. 4. This line represents the average non-*Tylonycteris* Vespertilionid and all points on it have the index 1 (or 100%). Each of the small horizontal bars represents the EI for a species. On the left, the range (vertical bars) and the mean (●) of the subfamilies are given, while on the right, the sequence of the means is scaled and the names of the units are given. — Figs. 7–9. Size indices (SIs) of cerebellum (Fig. 7), striatum (Fig. 8), and olfactory bulb (Fig. 9) of the four Vespertilionid subfamilies. The SIs in Fig. 8 are numerical values of the distances from the reference line in Fig. 5. For further explanations see Fig. 6

In *Vespertilioninae*, the average encephalization index is 95 with, and 96 without the *Tylonycteris* species. The encephalization of the two *Tylonycteris* species is 59 for *T. pachypus* and 62 for *T. robustula*. The non-*Tylonycteris* species range from 72 in *Glossophaga tylops* to 124 in *Myotis nattereri* (for EIs of single species see Table 1). Thus, there is a large variability (CV 14.9). The Vespertilionine subfamily, which is by far the largest, appears fairly heterogeneous in encephalization (Fig. 6). This is discussed in more detail in Part II.

In *Miniopterinae*, several groups in our collection were determined to be subspecies of *M. schreibersi*. These 'subspecies', however, behave like species with respect to their encephalization. Within a species, the (intraspecific) slope of the BrW/BoW regression line is generally distinctly lower (0.2–0.3, see STEPHAN et al. 1986) than the interspecific line (0.686, see above). Consequently, when relating subspecies with different BoWs to the interspecific regression line, the smaller subspecies would have higher EIs than the larger ones. In the material available, such differences do not exist. On the contrary, the two larger 'subspecies' have slightly larger indices than the two smaller ones (111 and 113 versus 109 and 108). The 8 species (or subspecies) of the *Miniopterinae*, which all belong to the genus *Miniopterus*, vary little in encephalization (CV 2.8). *Miniopterinae* appear to be largely homogeneous in relative brain size (Fig. 6). The average EI is 111, which is the second highest of the Vespertilionid subfamilies available.

In *Kerivoulinae*, the average EI is 130, which is clearly the highest of all subfamilies. The range is from 121 for *K. phalaena* to 140 for *K. papillosa*. Only *Myotis nattereri* reaches into this range.

In *Nyctophilinae*, the average EI is 102, which is the second lowest of the subfamily averages. The range is from 95 to 109.

Except for a few species of *Vespertilioninae*, which reach into the range of *Miniopterinae* and *Kerivoulinae*, there is only a small overlap between the poorly encephalized *Vespertilioninae* and *Nyctophilinae* and the *Miniopterinae* (intermediate EIs) and *Kerivoulinae* (high EIs) (Fig. 6). The average EIs of the *Vespertilioninae* are significantly different ($P < .01$) from those of the *Miniopterinae* and *Kerivoulinae*, but not from those of the *Nyctophilinae* (Table 5). The average EIs of the *Miniopterinae* are significantly different from those of the *Kerivoulinae* ($P < .01$) and from those of the *Nyctophilinae* ($P < .05$). No significant differences were found between the EIs of the *Nyctophilinae* and those of the *Kerivoulinae*. However, the two subfamilies are represented by only two species each.

Composition of the brains

The size indices (SIs) of the various species are given in Table 3, averages for subfamilies in Table 4, and significance tests in Table 5. The SIs of the net brain volumes (NBV) are similar to the encephalization indices. Differences are mainly due to the larger material in the encephalization studies. The various brain parts, however, may behave differently in the various subfamilies.

In *Vespertilioninae*, the average SIs of all the measured brain components are lower than 100, i. e., lower than in the average of the non-*Tylonycteris* Vespertilionids (Fig. 10). In most components, *Vespertilioninae* have the lowest average SIs, except for OBL, where *Nyctophilinae* are lowest, for MES, where *Miniopterinae* are lowest, and for BOL, PAL and SEP, where *Kerivoulinae* are lowest. Significant differences with *Miniopterinae* were found for CER, STR, NEO ($P < .01$) and TEL ($P < .05$), with *Kerivoulinae* for OBL, CER, TEL, STR, HIP, NEO ($P < .01$) and DIE ($P < .05$). No significant differences in any of the structures were found between *Vespertilioninae* and *Nyctophilinae* (Table 5). The Vespertilionine subfamily is by far the largest of the subfamilies, and for

Table 2

Volumes of the various brain parts in mm³ (n = number of specimens)

Code	Subfamilies and species	n	ENV	ORL	CER	MES	DIE	TEL	BOL	PAL	SEP	STR	SCH	HIP	NEO
Vespertilioninae															
2377	<i>Myotis adversus</i>	2	239.3	35.5	44.2	26.2	19.9	113.5	5.6	25.1	3.4	14.0	7.4	15.3	46.7
2399	<i>bechsteinii</i>	1	235.2	35.3	41.6	22.6	21.0	114.7	7.2	22.9	5.0	12.7	9.2	23.6	34.1
2403	<i>bocagei</i>	3	188.6	26.1	35.5	21.1	16.6	89.3	4.3	17.9	2.7	10.3	6.0	10.2	37.9
2419	<i>dasycneme</i>	1	291.9	46.3	50.9	34.8	25.3	134.6	7.7	27.5	4.6	13.5	11.5	25.2	44.6
2489	<i>myotis</i>	2	452.0	70.8	86.8	46.0	42.1	206.3	11.5	36.7	6.7	22.6	17.8	34.4	74.6
2495	<i>nattereri</i>	1	207.9	30.1	37.5	20.0	17.4	102.9	4.7	21.1	3.4	10.0	8.5	21.5	33.7
2497	<i>nigricans</i>	3	128.1	17.0	23.6	14.6	9.9	62.8	2.9	14.2	2.0	7.9	4.0	9.0	22.8
2591	<i>Pipistrellus erasmeeus</i>	3	125.2	19.9	21.1	14.6	12.0	57.6	3.6	12.9	1.9	5.7	4.0	6.2	21.3
2601	<i>imbricatus</i>	2	129.2	21.6	20.3	16.3	13.7	57.3	3.3	12.1	2.2	6.2	3.1	9.2	21.2
2631	<i>nanus</i>	3	94.5	15.2	18.4	10.2	7.9	42.8	2.0	9.3	1.4	5.2	2.9	6.3	15.6
2672	<i>Nyctalus stenopterus</i>	2	234.5	38.2	41.6	27.1	22.5	105.1	6.2	23.5	3.2	11.5	6.1	14.2	40.4
2675	<i>Olaschropterus tylopsus</i>	3	92.2	16.0	17.7	11.0	7.6	39.9	2.2	9.7	1.3	4.4	2.4	6.3	14.0
2681	<i>Eptesicus brasiliensis</i>	3	189.0	28.1	36.1	21.6	15.3	89.9	4.5	19.8	3.0	9.7	6.2	11.9	34.8
2695	<i>flavescens</i>	2	179.5	29.9	34.2	21.1	15.1	79.2	4.8	16.5	2.4	8.1	5.2	10.7	31.5
2699	<i>fuscus</i>	1	226.5	31.4	39.0	22.8	19.2	114.1	6.2	20.1	3.4	11.0	9.9	19.9	43.6
2723	<i>pumilus</i>	2	115.2	19.9	21.5	12.7	9.7	52.4	2.6	10.6	1.9	5.5	3.8	9.7	18.3
2773	<i>Tylonycteris puchypus</i>	4	69.8	13.7	10.8	9.0	6.2	30.1	1.6	7.0	1.0	2.9	2.1	4.1	11.4
2775	<i>robustula</i>	3	113.2	21.6	21.5	13.1	9.6	47.4	3.0	10.8	1.5	4.4	3.8	6.5	17.4
2796	<i>Claudycteris humeralis</i>	2	136.4	20.1	31.6	16.3	10.3	58.1	1.5	10.2	1.4	6.7	4.0	10.2	24.1
2799	<i>poensis</i>	1	194.4	30.3	43.3	22.9	15.2	82.7	2.4	15.9	2.0	8.3	5.9	12.2	36.0
2807	<i>Chalinolobus aoudi</i>	2	224.8	33.7	49.9	23.6	17.0	100.6	5.7	19.3	3.4	10.2	7.1	17.6	37.3
2809	<i>morio</i>	2	174.9	28.2	35.0	19.4	13.7	78.6	3.2	12.1	2.7	8.1	6.3	15.0	28.2
2831	<i>Nycticeius balstoni</i>	1	149.1	23.6	27.0	16.7	13.8	68.0	4.1	14.4	2.0	6.6	5.0	10.2	25.7
2843	<i>Rhogeessa parvula</i>	3	104.2	16.9	17.9	12.7	8.9	47.8	2.8	10.7	1.6	5.2	3.7	6.3	17.5
2857	<i>Scotophilus dinganii</i>	3	415.1	58.6	82.1	45.8	35.2	193.4	11.1	35.9	4.5	18.6	12.7	22.8	87.8
2861	<i>kuhli</i>	3	310.3	47.2	56.3	35.4	27.8	142.6	6.5	27.0	4.0	13.0	9.5	18.0	64.6
2877	<i>Lasius borealis</i>	1	161.4	23.4	41.0	18.7	12.5	65.8	4.0	12.0	2.1	6.3	4.6	9.0	27.8
Miniopterinae															
2907	<i>Miniopterus australis</i>	2	194.9	27.3	39.8	18.1	14.7	95.0	4.5	17.2	2.7	11.7	6.1	15.1	37.7
2911	<i>inflatus</i>	3	310.7	44.2	80.2	27.7	22.1	136.5	6.9	24.2	4.0	16.3	9.4	19.1	56.4
2913	<i>medius</i>	3	204.7	28.6	47.1	18.2	15.8	95.0	4.0	18.4	2.6	11.5	6.3	15.0	37.2
2920	<i>M. schreib. arenarius</i>	3	227.5	31.4	49.4	20.4	17.0	109.3	5.3	20.3	2.9	13.0	7.7	16.5	43.6
2921	<i>M. blopetis</i>	2	232.5	41.4	73.7	26.7	20.5	130.2	6.1	25.4	3.7	16.1	9.1	19.8	50.0
Kerivoulinae															
2967	<i>Kerivoula papillosa</i>	2	295.7	35.3	60.5	24.3	21.9	153.7	3.8	17.7	3.0	12.9	6.4	36.0	73.9
2971	<i>phalaena</i>	1	110.3	17.2	23.2	13.0	9.1	47.8	1.6	8.1	1.1	7.1	3.0	6.8	20.1
Nyctophilinae															
2999	<i>Nyctophilus geoffroyi</i>	2	156.9	21.6	29.2	15.4	12.6	78.1	4.5	13.6	2.1	8.1	5.7	12.3	31.8
3005	<i>tiniorensis</i>	2	240.6	31.3	43.9	26.7	19.2	119.5	5.7	22.4	3.0	12.9	8.5	19.4	47.6

the SIs of all structures (except HIP), the most variable. More detailed information on brain differences between the various genera is given in Part II.

In *Miniopterinae*, the average SIs of all structures are higher than 100, except that of MES (95), which is the lowest of the subfamilies (Fig. 10). For PAL and SEP, the highest average SIs in the subfamilies were found in *Miniopterinae*. For CER and STR, *Miniopterinae* also have relatively high SIs. In encephalization the species (or 'subspecies', see above) varied only slightly. This applies to most structures as well, and in just two of them (BOL and HIP) was the CV higher than 10. Based on the present material, this subfamily

Table 3

Size indices (SI; $\times 100$) of the various brain parts (n = number of specimens)

Code	Subfamilies and species	n	NEV	OBL	CER	MES	DIE	TEL	BOL	PAL	SEP	STR	SCH	MIP	NEO
Vespertilioninae															
2377	<i>Nyctotis adavatus</i>	2	106	106	98	110	107	109	104	114	107	124	98	81	117
2399	<i>Neotoma bechsteini</i>	1	114	115	101	102	122	120	148	123	170	121	134	157	93
2403	<i>Neotoma bocagii</i>	3	113	105	107	116	120	116	115	110	113	119	110	85	128
2419	<i>Nyctinomys danyanensis</i>	1	102	110	89	118	107	102	109	108	112	96	118	120	87
2459	<i>Nyctinomys myotis</i>	2	101	107	96	103	112	99	95	98	102	108	111	103	93
2495	<i>Nyctinomys nattereri</i>	1	124	120	113	110	127	134	125	139	143	116	137	179	114
2497	<i>Nyctinomys nigricans</i>	3	103	96	102	111	103	115	116	131	121	124	110	106	109
2591	<i>Pipistrellus crassulus</i>	2	85	90	72	90	99	84	110	96	92	74	84	77	81
2601	<i>Pipistrellus fabricatus</i>	2	85	96	68	98	110	82	98	88	103	78	64	85	79
2631	<i>Nyctinomys nanus</i>	3	87	93	86	83	89	86	89	93	94	90	86	91	82
2672	<i>Nyctinomys stenopterus</i>	2	84	92	74	93	97	81	90	94	79	84	63	70	81
2675	<i>Glaucidium tylops</i>	3	71	83	70	76	72	67	81	79	68	64	60	68	61
2681	<i>Eptesicus brasiliensis</i>	3	97	90	94	104	95	100	100	113	109	98	97	84	101
2695	<i>Eptesicus flavescens</i>	2	84	94	80	93	85	80	95	85	79	75	74	69	83
2699	<i>Eptesicus fuscus</i>	1	89	83	76	85	90	96	100	88	93	87	115	107	96
2723	<i>Eptesicus pumilus</i>	2	84	91	79	83	96	83	88	84	96	76	88	98	75
2773	<i>Tylonycteris pachypus</i>	4	58	76	46	67	63	55	65	63	63	45	56	47	54
2775	<i>Tylonycteris robustula</i>	3	63	80	60	67	64	57	73	56	59	48	64	50	54
2796	<i>Glaucidium humeralis</i>	2	99	97	116	107	91	92	51	81	70	92	92	103	99
2799	<i>Glaucidium yomae</i>	1	104	109	117	114	96	96	56	94	75	87	97	90	108
2807	<i>Chalinolobus gouldi</i>	2	98	98	109	97	89	94	104	93	102	88	92	105	91
2809	<i>Chalinolobus morio</i>	2	95	103	96	96	90	93	76	73	104	86	106	135	87
2831	<i>Nyctinomys baistoni</i>	1	77	82	70	80	86	76	91	82	72	66	79	72	74
2843	<i>Eptesicus parvula</i>	3	91	98	80	98	95	91	119	102	102	84	103	77	87
2857	<i>Scotophilus dinganii</i>	3	110	105	107	119	111	110	112	107	82	103	95	81	129
2861	<i>Scotophilus kuhli</i>	3	93	96	84	107	100	92	77	91	83	81	82	73	109
2877	<i>Lasiurus borealis</i>	1	92	90	118	99	87	82	101	76	86	70	81	71	90
Miniopterinae															
2907	<i>Miniopterus australis</i>	2	119	111	123	101	108	126	124	115	118	138	115	128	130
2911	<i>Miniopterus inflatus</i>	3	113	106	145	96	96	107	102	99	100	122	100	95	115
2913	<i>Miniopterus medius</i>	3	110	103	127	90	102	110	95	109	100	120	103	111	112
2920	<i>M. schreib. aeneus</i>	3	108	101	118	91	96	113	107	108	99	122	110	108	117
2921	<i>M. schreib. a. bleptotis</i>	2	111	106	140	97	94	107	95	108	98	124	103	103	106
Kerivoulinae															
2967	<i>Kerivoula papillosa</i>	2	142	114	146	109	127	139	77	94	100	122	93	238	200
2971	<i>Kerivoula phalaena</i>	1	121	126	130	125	122	114	89	96	86	141	110	105	126
Nyctophilinae															
2999	<i>Nyctophilus geoffroyi</i>	2	95	87	89	86	92	102	122	90	91	94	106	103	109
3005	<i>Nyctophilus timoriensis</i>	2	110	96	100	114	106	118	110	113	97	117	116	121	122

appears to be fairly homogeneous. Significant differences with Kerivoulinae were found for MES, DIE ($P < .01$), OBL and PAL ($P < .05$), with Nyctophilinae for OBL and CER ($P < .05$) (Table 5). For significant differences with Vespertilioninae, see previous paragraph.

In *Kerivoulinae*, the average SIs of many structural complexes are highest of the subfamilies, except those of BOL, PAL, SEP and SCH (Fig. 10). In the olfactory structures (BOL, PAL) and the septum (SEP), the Kerivoulinae in fact have the lowest average SIs. Thus, the various brain parts may behave very differently: the SIs range from 83 for BOL

to 171 for HIP. Next to HIP, the Kerivoulinae have relatively large average SIs for NEO (163), CER (138) and STR (131) (Figs. 7, 8, 10). The brain composition of the two species is similar for most structures, but highly different for NEO (CV 32.1) and especially for HIP (CV 54.8). In both structures, *K. papillosa* has distinctly higher SIs than *K. phalaena* (1.59 times in NEO, 2.27 times in HIP) (Fig. 11). The very large size of the HIP is paralleled by a unique structural differentiation (see Part III). Significant differences with Vespertilioninae and Miniopterinae were described in the previous paragraphs. With Nyctophilinae, they were found for CER only ($P < .05$) (Table 5). They may exist in other structures also, but cannot be demonstrated because of the small number of species in both subfamilies.

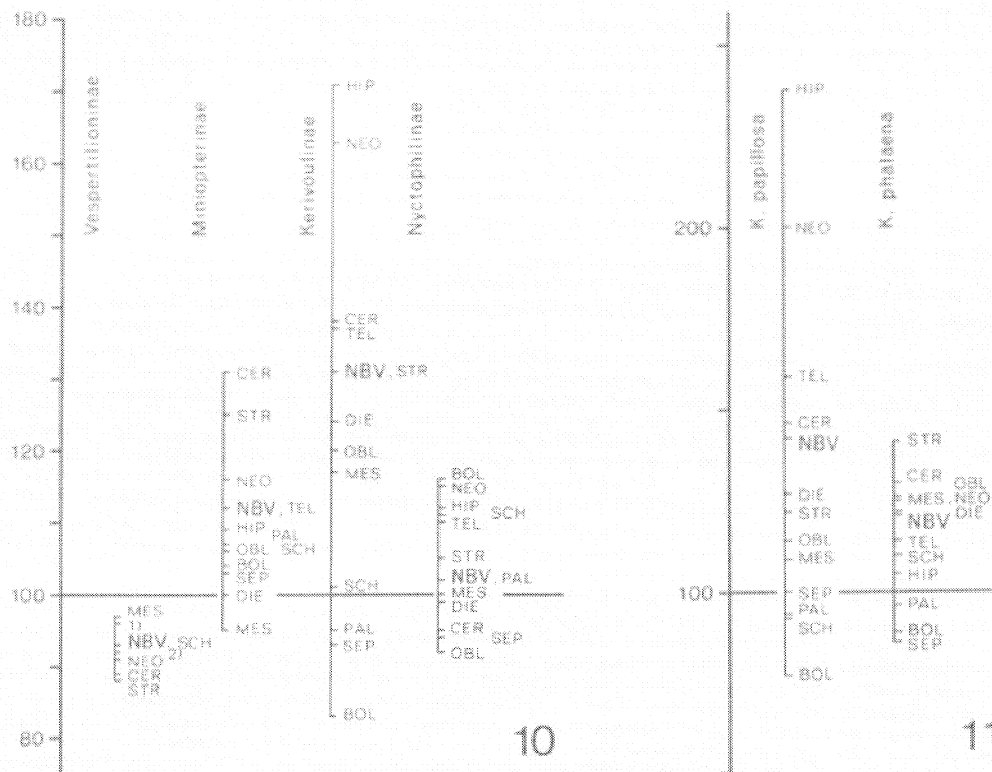
In *Nyctophilinae*, the highest average SIs in the subfamilies were found for BOL and SCH, the lowest for OBL (Figs. 9, 10). The average SIs range from 92 for OBL to 116 for BOL. The brain composition of the two species is similar; the highest CV (19.8) was found for MES. Based on the weak material available, this subfamily appears to be fairly homogeneous and there are nearly no significant differences from other subfamilies (see previous paragraphs).

The sequence found in the average EIs and in the average SIs for the total brain (i.e., lowest values in Vespertilioninae, slightly higher values in Nyctophilinae, intermediate values in Miniopterinae and high values in Kerivoulinae) is modified in various ways and intensities for the average SIs of many of the structures. Thus, the lowest averages for OBL and SEP were found in Nyctophilinae, the lowest averages for BOL, PAL and SEP in Kerivoulinae, and the lowest average for MES in Miniopterinae. The highest averages for BOL and SCH were found in Nyctophilinae, and the highest averages for PAL and SEP in Miniopterinae. Most of the differences are quite small, but there are some significant differences between the subfamilies, such as the high SIs of CER and STR in Minio-

Table 4

Average indices ($\times 100$) and coefficients of variation (CV, underlined) of Vespertilionid subfamilies
Average EI and SIs of non-*Tylonycteris* Vespertilionids = 100. (N = number of species)

	N	EI	N	Size indices and CVs						
				NEV	OBL	CER	MES	DIE	TEL	
Vespertilionidae	58	99	36	98	99	98	98	98	99	
		<u>15.5</u>		<u>17.6</u>	<u>11.7</u>	<u>24.8</u>	<u>14.3</u>	<u>15.6</u>	<u>21.2</u>	
Vespertilioninae	46	95	27	93	96	89	97	96	92	
		<u>14.9</u>		<u>16.5</u>	<u>11.3</u>	<u>21.2</u>	<u>15.0</u>	<u>16.5</u>	<u>19.7</u>	
Miniopterinae	8	111	5	112	106	131	95	100	112	
		<u>2.8</u>		<u>3.7</u>	<u>3.7</u>	<u>8.8</u>	<u>4.8</u>	<u>5.6</u>	<u>7.0</u>	
Kerivoulinae	2	130	2	131	120	138	117	124	137	
		<u>10.3</u>		<u>11.3</u>	<u>7.1</u>	<u>8.2</u>	<u>9.7</u>	<u>2.8</u>	<u>23.3</u>	
Nyctophilinae	2	102	2	102	92	95	100	99	110	
		<u>9.7</u>		<u>10.3</u>	<u>7.0</u>	<u>8.2</u>	<u>19.8</u>	<u>10.0</u>	<u>10.3</u>	
				Size indices and CVs						
				BOL	PAL	SEP	STR	SCH	HIP	NEO
Vespertilionidae				97	98	96	97	96	100	100
		<u>20.6</u>		<u>17.6</u>	<u>22.2</u>	<u>25.3</u>	<u>22.0</u>	<u>36.1</u>	<u>26.6</u>	
Vespertilioninae		96		96	96	88	93	92	91	
		<u>22.3</u>		<u>19.8</u>	<u>25.5</u>	<u>24.1</u>	<u>25.0</u>	<u>32.4</u>	<u>21.4</u>	
Miniopterinae		104		107	103	125	106	109	116	
		<u>11.4</u>		<u>5.7</u>	<u>8.2</u>	<u>5.8</u>	<u>5.8</u>	<u>11.2</u>	<u>7.5</u>	
Kerivoulinae		83		95	93	131	101	171	163	
		<u>10.2</u>		<u>1.5</u>	<u>10.6</u>	<u>10.2</u>	<u>11.8</u>	<u>54.8</u>	<u>32.1</u>	
Nyctophilinae		116		102	94	105	111	112	115	
		<u>7.3</u>		<u>16.0</u>	<u>4.5</u>	<u>15.4</u>	<u>6.4</u>	<u>11.4</u>	<u>8.0</u>	



Figs. 10-11. SI profiles of the four subfamilies (Fig. 10) and SIs of the two *Kerivoula* species (Fig. 11). The SIs are given in percentage form (i.e., $\times 100$). In Vespertilioninae (left column in Fig. 10) five structures (OBL, DIE, BOL, PAL, SEP) having the same SI are marked by 1, two structures (TEL, HIP) by 2. For each structure, the average of the 34 non-*Tylonycteris* Vespertilionid species is 100. (For abbreviations see Material and methods)

Table 5

Significance tests (t-tests) for differences in brains and brain parts between the various subfamilies

(** $P < .01$; * $P < .05$; ns not significant)

Compared subfamilies	EL	NBV	OBL	CER	MES	DIE	TEL
Vespertilioninae versus							
Miniopterinae	**	*	ns	**	ns	ns	*
Kerivoulinae	**	**	**	**	ns	*	**
Nyctophilinae	ns	ns	ns	ns	ns	ns	ns
Miniopterinae versus							
Kerivoulinae	**	*	*	ns	**	**	ns
Nyctophilinae	*	ns	*	*	ns	ns	ns
Kerivoulinae versus							
Nyctophilinae	ns	ns	ns	*	ns	ns	ns
	BOL	PAL	SEP	STR	SCH	HIP	NEO
Vespertilioninae versus							
Miniopterinae	ns	ns	ns	**	ns	ns	**
Kerivoulinae	ns	ns	ns	**	ns	**	**
Nyctophilinae	ns	ns	ns	ns	ns	ns	ns
Miniopterinae versus							
Kerivoulinae	ns	*	ns	ns	ns	ns	ns
Nyctophilinae	ns	ns	ns	ns	ns	ns	ns
Kerivoulinae versus							
Nyctophilinae	ns	ns	ns	ns	ns	ns	ns

pterinae and Kerivoulinae and the high SIs of NEO and HIP in Kerivoulinae. The very high average for NEO, however, is mainly due to *K. papillosa*, that of HIP is due solely to this species, whereas the HIP of *K. phalaena* (SI = 105) is not far from the overall average (Fig. 11).

The range of the average SIs of the brain structures differs according to subfamily. This diversification can be expressed by maximum/minimum ratios in the four subfamilies listed in Table 4. The lowest max/min ratio (1.10) was found in Vespertilioninae; this is not surprising since it is by far the largest of the subfamilies and thus has a large influence on the position of the reference baselines. In ascending sequence follow Nyctophilinae (1.26), Miniopterinae (1.38) and Kerivoulinae (2.06). The last of these has by far the highest diversity in the size of the various structures (Fig. 10).

Percentage values are less instructive for interspecific comparisons, since 1. intracerebral percentages may be similar even if the brain size in isoponderous species is quite different, and 2. changing percentages of one structure inevitably cause changes in the percentages of all other structures, which are also included in the reference system, as e.g., in NBV or TEL. For example: the very large hippocampus and neocortex of *K. papillosa* produce, in the percentage composition of TEL, low values for the other telencephalic structures even when the size indices indicate a clear increase, as e.g., for the striatum (see Fig. 8).

Discussion

When brain size was related to body size using allometric methods, it was found that several groups of Insectivora (Tenrecinae, Soricidae, Erinaceidae; = 'basal' Insectivora) have the lowest relative brain size (expressed in form of EIs). The composition of the brains is similar in these groups and, compared with other mammalian forms, is characterized by relatively large medulla oblongata, mesencephalon and olfactory structures and relatively small neocortex, striatum, cerebellum and diencephalon. The brain characteristics of these Insectivora must be regarded as the most conservative (plesiomorph) of recent mammals. Great distances from this conservative stage were reached by Primates, and by far the greatest distance by man. Quantitative investigations in a series of Primates have demonstrated that striatum, diencephalon, cerebellum and neocortex are strongly enlarged, with an extremely high increase of the neocortex (STEPHAN and ANDY 1969; STEPHAN 1972). The larger size of these brain parts is derivative (apomorph).

In Chiroptera, the relative size of neocortex and striatum is distinctly greater in the more highly encephalized Megachiroptera than in Vespertilionids (STEPHAN 1985). With respect to encephalization and many characteristics of brain composition, Vespertilionids are the most conservative of the Chiroptera. The strong reduction of their olfactory structures (STEPHAN and PIRLOT 1970; PIRLOT and POTTIER 1977), however, is probably derivative. Of the various Vespertilionid subfamilies, the highest encephalization was found in Kerivoulinae, and an analysis of the brain composition has shown that this is mainly due to enlargement of hippocampus, neocortex, cerebellum and striatum. Kerivoulinae seem to have the greatest number of apomorph brain characteristics of the Vespertilionids. Similarly, in Miniopterinae, i.e., in the subfamily with the second highest EI, cerebellum, striatum and neocortex also are relatively large.

According to the number of derivative brain characteristics (encephalization, composition, cover of mesencephalon), and the intensity of derivation, the following ascending sequence is obtained: Vespertilioninae, Nyctophilinae, Miniopterinae, and Kerivoulinae. This is not paralleled by skull and teeth characteristics. Based on TATE (1942), KINGDON (1974, p. 262) stated that "Kerivoulinae and Miniopterinae are less progressive subfamilies", and "The more advanced (Vespertilionid) genera have shortened the face and reduced the number of teeth", and "Some genera, particularly the most advanced, have

radiated into a large number of species, ... *Eptesicus* and *Pipistrellus*". Thus, in Vespertilionids, apomorph and plesiomorph characteristics are mixed. This is not surprising since the characteristics may change independently in adaptive radiations. Number and structure of teeth depend to a great extent on the type of food and the prevailing feeding strategies. This accounts for several parts of the brain too, as e.g., the olfactory structures, which are especially small in aerial insectivorous bats and large in frugivorous species. Other brain structures, as e.g., neocortex, appear to be less closely related in their overall size to the various kinds of specializations. In adaptive radiations, the neocortex appears to develop nearly always in the direction of an enlargement. A few possible exceptions to this trend are discussed in Part II. The general trend may make neocortex size the best criterion for determining the distances between derived forms and conservative ones.

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Zusammenfassung

Gehirne von Vespertilioniden. I. Charakteristika der Unterfamilien

Makromorphologie der Gehirne und Encephalisation wurden bei 58 Vespertilioniden-Arten verglichen, die Hirnzusammensetzung bei 36 Arten: 46 bzw. 27 Arten von Vespertilioninae, 8 bzw. 5 Arten von Miniopterinae, 2 Arten von Kerivoulinae und 2 Arten von Nyctophilinae. Unterschiede zwischen den Unterfamilien finden sich in der Bedeckung des Mittelhirns. Es ist nahezu komplett verdeckt bei *Kerivoula papillosa* (Kerivoulinae), mehr als zur Hälfte bedeckt (durch das Cerebellum) bei den Miniopterinae und kaum oder nicht bedeckt bei Nyctophilinae und Vespertilioninae. In der Encephalisation liegen die Kerivoulinae am höchsten (mittlerer EI 130), gefolgt von Miniopterinae (111), Nyctophilinae (102) und Vespertilioninae (95). Die höhere Encephalisation der Kerivoulinae und Miniopterinae ist begleitet von einer deutlichen Zunahme der relativen Größe von Cerebellum und Striatum und bei Kerivoulinae auch von Hippocampus und Neocortex.

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BUCHBESPRECHUNG

MANSON, D. C. M.: *Eriophyoidea except Eriophyinae (Arachnida: Acari)*. Fauna of New Zealand 4, NZ \$ 10.50; *Eriophyinae*. Fauna of New Zealand 5, NZ \$ 9. ORDISH, R. G.: *Hydraenidae (Insecta: Coleoptera)*. Fauna of New Zealand 6, NZ \$ 7.50. Wellington: Science Information Publishing Center, DSIR. 1984.

The "Fauna of New Zealand" series of manuals dealing with the country's non-marine invertebrates was launched in late 1982, and the release of another three volumes in late 1984 testifies to an encouraging vigour of this ambitious and important project.

The series editors, based in the systematics group of the DSIR Entomology Division are to be commended for the design of these manuals. Introductory chapters outline the history and current status of New Zealand research on the group in question and provide comprehensive accounts of the animals' structure and biology; then follows the principal treatment including keys and descriptions, and the volumes are concluded by reference lists, indices and very rich figure sections, featuring habitus illustrations (those in the hydraenid volume, by DSIR artist D. HELMORE are particularly outstanding) as well as diagrams of diagnostic details. The books are printed on good quality paper which allows reproduction of half-tone illustrations, but apart from a few SEM pictures of stridulatory spines in the hydraenid volume all pictures in the present parts are line drawings. The text is offset reproduced from typescript and the price of the volumes thereby kept within manageable limits; a laudable policy.

The two-volume treatment of eriophyoid mites covers 37 genera with a total of 109 species; 14 of the genera and 54 of the species are new. Serious study of the eriophyoid mites associated with native New Zealand plants began only about the middle of this century, and MANSON's expectations (vol. 4, p. 8) that "many more species await discovery" will undoubtedly be fulfilled. In addition to keys and descriptions the treatment contains a host-plant list which will prove invaluable to field workers; it seems a pity, however, that no illustrations are provided of the characteristic plant malformations caused by several of these mites.

ORDISH's treatment of the small, mostly aquatic hydraenid beetles covers even more novel ground: of the five genera two are new and of the 32 species no less than 27 are new. Many of the new species are documented from very extensive series, but "further collecting is needed before the ranges of some can be delimited with any confidence" (p. 7) and no larval/adult associations have so far been established in mainland New Zealand.

There can be no doubt that these books will admirably serve their primary purpose, viz. to enable workers to identify New Zealand collections of the taxa treated. But their potential importance goes far beyond this, because every attempt of placing elements of New Zealand biota into their proper position in "nature's own hierarchy" is a contribution towards the ultimate understanding of what remains one of the major puzzles of biogeography: "explain New Zealand and the world falls into place around it", as one biogeographer put it some years ago (NELSON, *Syst. Zool.* 24: 494, 1975). With this in mind the phylogeny discussions offered in the present volumes are admittedly somewhat disappointing. At least as far as descriptions of new endemic genera are concerned, I believe that editorial demands for the depth of the accompanying comparative discussions should be stiffened in forthcoming parts of this series.

KRISTENSEN, Copenhagen

